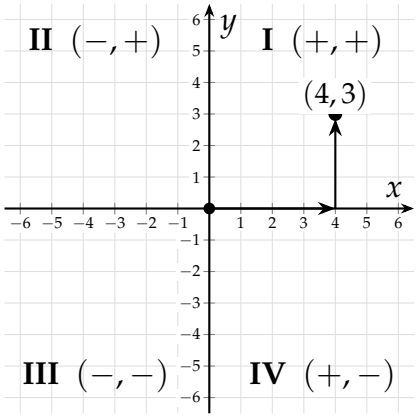
Graphing on the Coordinate Plane

Test review: points, quadrants, reflections, distance, shapes and proportional graphs

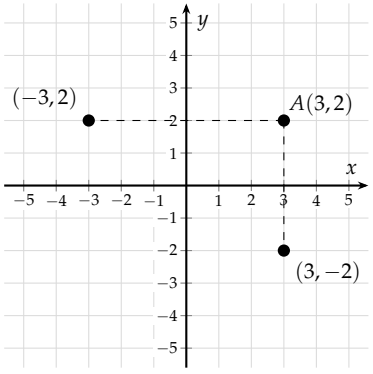
1 The coordinate plane



- The x**-axis** goes left–right; the y**-axis** goes up–down. They cross at the **origin** $(0, 0)$.
- An **ordered pair** (x, y): **first** walk left/right (x), **then** up/down (y).
  "Walk to the elevator, then ride it."
- The axes cut the plane into four **quadrants**, numbered I–IV **counterclockwise**, starting at the top right.
- A point **on an axis** (like $(0, 5)$ or $(-2, 0)$) is in **no** quadrant.

Watch out: $(4, 3)$ and $(3, 4)$ are **different** points. Order matters: x always comes first.

2 Reflections — flip over an axis



Reflect over…	Rule	Example
the x-axis	$(x, y) \to (x, -y)$	$(3, 2) \to (3, -2)$
the y-axis	$(x, y) \to (-x, y)$	$(3, 2) \to (-3, 2)$

Memory trick: over the x**-axis**, the x **stays**; only y changes sign. Over the y**-axis**, the y **stays**.

3 Distance between two points on a line

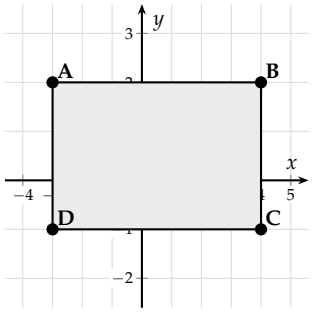
Same x or same y? Count — or use absolute value

Same quadrant / same side: subtract. $(2,5)$ to $(2,1)$: $5-1=4$ units.
Opposite sides of an axis: add the absolute values. $(-3,4)$ to $(5,4)$: $|-3|+|5| = 3+5=8$ units.
One rule for both: **distance** $=$ | difference |. $|5-(-3)| = |8| = 8$. Distance is never negative.

Simple case first: from $(-1,0)$ to $(1,0)$ — count the squares: 2. And $|-1|+|1|=2$ ✓. That's why you **add** across the axis.

4 Shapes on the coordinate plane

Example · rectangle $ABCD$ with $A(-3,2)$, $B(4,2)$, $C(4,-1)$, $D(-3,-1)$

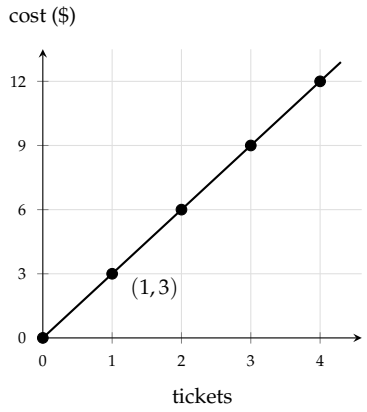


Length AB: same y, so $|4-(-3)|=7$.
Width BC: same x, so $|2-(-1)|=3$.
Perimeter $=7+3+7+3=20$ units.
Area $=7\times 3=21$ square units.

5 Graphing a proportional relationship

Example · Tickets cost $3 each

tickets x	0	1	2	3	4
cost y ($)	0	3	6	9	12



How to tell it's proportional from a graph:

1. the points are on a **straight line**, and
2. the line goes **through the origin** $(0,0)$.

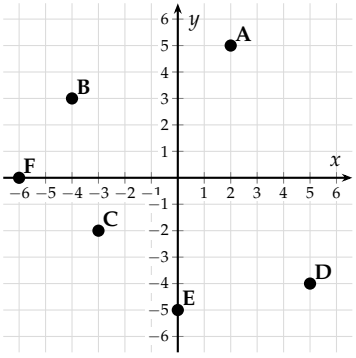
The **constant of proportionality** (unit rate) is $k = \frac{y}{x} = \frac{3}{1} = 3$. Equation: $y = 3x$.
Look for the point $(1, k)$: here $(1,3)$ means *1 ticket costs $3*.
A point like $(4,12)$ means *4 tickets cost $12*.

Not proportional: a straight line that **doesn't** pass through $(0,0)$ — for example a $5 entry fee plus $3 per ride: $y = 3x + 5$ starts at $(0,5)$.

6 Practice

Show your work. Label every point you plot.

A. Read and plot points EASY



1. Write the coordinates of each point.

A _____ B _____
C _____ D _____
E _____ F _____

2. Which quadrant is each of A, B, C, D in? Which points are on an axis?

3. Plot and label on the same grid:
$P(3,1)$, $Q(-5,-4)$, $R(0,4)$, $S(4,-2)$.

B. Signs and quadrants EASY MEDIUM

4. Name the quadrant: (a) $(-7, 2)$ (b) $(6, -1)$ (c) $(-0.5, -8)$ (d) $(3\frac{1}{2}, 4)$

5. A point has a negative x-coordinate and a positive y-coordinate. Which quadrant is it in?

6. True or false: the point $(0, -3)$ is in Quadrant IV. Explain.

C. Reflections MEDIUM

7. Reflect $(5, -2)$ over the x-axis. Then reflect $(5, -2)$ over the y-axis.

8. $(-4, 6)$ is reflected over the y-axis, and then the new point is reflected over the x-axis. Where does it end up? Which quadrant?

9. Point M is (a, b) and point N is $(a, -b)$. Over which axis is N the reflection of M?

D. Distance MEDIUM

10. Find the distance between $(3, 7)$ and $(3, -2)$.

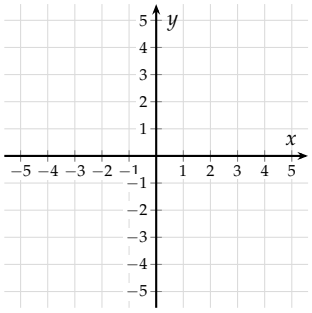
11. Find the distance between $(-6, -1)$ and $(-2, -1)$.

12. Point A is at $(-5, 4)$. Point B is 9 units to the right of A. Find B. Which quadrant is B in?

13. A diver is at -12 m and a seagull is at $+15$ m, straight above the diver. How far apart are they?

E. Shapes MEDIUM HARD

14. Plot $J(-4,3)$, $K(2,3)$, $L(2,-2)$ and $M(-4,-2)$ and connect them in order.
(a) What shape is $JKLM$?
(b) Find its perimeter.
(c) Find its area.



15. Three corners of a square are $(-1,4)$, $(3,4)$ and $(3,0)$. Find the fourth corner and the area of the square.

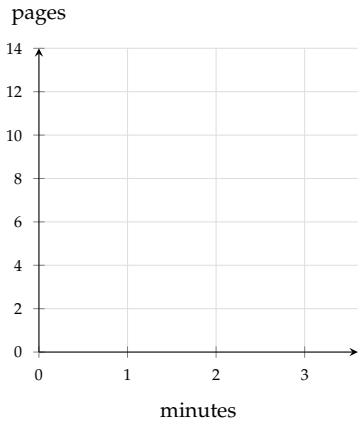
16. HARD A triangle has vertices $(-2,-3)$, $(4,-3)$ and $(1,5)$. Find its area.

F. Proportional relationships MEDIUM HARD

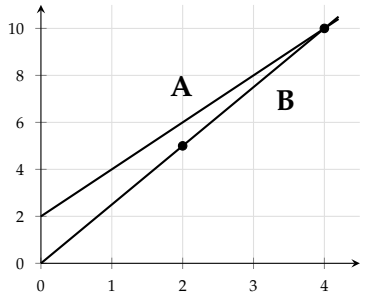
17. A printer prints 4 pages per minute.

minutes x	0	1	2	3
pages y				

(a) Complete the table and graph it.
(b) Is the relationship proportional? How can you tell from the graph?
(c) Write an equation. (d) What does the point $(1,4)$ mean?



18. Which graph shows a proportional relationship? For that one, find the constant of proportionality k.



19. HARD A proportional graph passes through $(6, 15)$. Find k, write the equation, and find y when $x = 10$.

G. Challenge HARD

20. Point $P(-3, a)$ is 7 units from $Q(-3, 2)$. Find **both** possible values of a.

21. The rectangle $(1, 1)$, $(5, 1)$, $(5, 3)$, $(1, 3)$ is reflected over the y-axis. List the new vertices. Did the area change?

Before the test: x first, then y · quadrants go counterclockwise from top right · across an axis, **add** distances · proportional = straight line **through the origin**, unit rate at $(1, k)$.